

SKILLS

---

- **Backend Engineering:** Python, FastAPI, TypeScript, Node.js, Go, SQL, REST APIs, GraphQL, microservices
- **AI/LLM Platforms:** LLMs, RAG, LangChain, LangGraph, tool calling, embeddings, prompt/context engineering, OpenAI API
- **AI Quality & Safety:** evaluation sets, regression testing, grounding controls, citation validation, guardrails, tracing, audit logs
- **Data Systems:** PostgreSQL, pgvector, Elasticsearch, Redis, Kafka, Airflow, event streams, schema design, caching
- **Cloud & Infrastructure:** Kubernetes, Docker, Terraform, CI/CD, AWS (EKS, Bedrock, Lambda, S3, Kinesis), GCP
- **Reliability & Observability:** OpenTelemetry, ELK, distributed tracing, SLOs, alerting, performance analysis, incident response
- **Technical Leadership:** system design, architecture reviews, roadmap ownership, cross-functional planning, mentoring, executive communication

EXPERIENCE

---

- **Royal Bank of Canada (RBC)** Toronto, Canada  
*Staff Software Developer & Enterprise Architect – AI Platform* Nov 2022 – Present
  - **ASB Assist – LLM Platform Capabilities (In Development):** Building a conversational AI layer that converts natural-language requirements into compliant architecture diagrams and service recommendations using LLM + RAG orchestration, retrieval, tool integration, context engineering, evaluations, security controls, and audit trails.
  - **LLM Reliability & Evaluation:** Designed grounding, context-assembly, citation-validation, and offline evaluation patterns for regulated AI workflows; compare prompt and retrieval changes, detect quality regressions, and preserve traceability before production rollout.
  - **0-to-1 Backend & Platform Ownership:** Led production delivery of a patent-pending full-stack platform using Node.js, TypeScript, React, and REST APIs; wrote 99.6% of production code and 100% of core logic for rules-based recommendations, real-time rendering, and auto-certification.
  - **Distributed Data Infrastructure at Scale:** Architected an OpenTelemetry platform using Vector, Fluent Bit, Kafka, and Logstash that ingests 150 TB/day across 10,000+ systems; standardized schemas and vendor-agnostic integrations adopted by 20+ engineering teams.
  - **Architecture, Delivery & Force Multiplication:** Lead 5 engineers through ambiguous multi-team initiatives, translating product and architecture goals into technical plans, ADRs, code/design reviews, production handoff, observability standards, mentoring, and roadmap decisions.
- **HealthCard (Acquired Startup)** Remote  
*Blockchain Architect & First Full Stack Engineer* Aug 2021 – Jul 2022
  - **Document Intelligence & Verification:** Integrated transformer-based document classification with insurance and credential workflows; deployed inference APIs, monitoring, trust scoring, and audit trails that reduced manual review by 60%.
  - **Founding-Engineer Production Delivery:** Built the product from scratch across React/TypeScript, Node.js/Express APIs, verification services, and AWS EKS with HIPAA-aligned data handling; the team scaled to 20+ engineers before acquisition.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)  
*IT Developer / Software Developer Co-op* Apr 2022 – Nov 2022
  - **Public-Scale Platform Modernization:** Migrated a federal tax platform from Apache Struts to React/TypeScript and contributed to API layers serving millions of Canadians with accessibility, security, maintainability, and automated quality gates.
- **Dalhousie University** Halifax, Canada  
*Research & Teaching Assistant* Jan 2021 – Aug 2022
  - **Python Systems Research & Cloud Teaching:** Built Python tracing tools for call-stack decision algorithms and supported Algorithms, Software Development, and graduate Advanced Cloud Computing courses.

PROJECTS

---

- **Production-Style Document AI Platform:** Built a Python/FastAPI + React RAG system using PostgreSQL/pgvector and Elasticsearch, with asynchronous ingestion, tenant isolation, evaluation sets, citation validation, tracing, and human review workflows.
- **Automatic Trading Algorithm:** Built a sub-second distributed system integrating live brokerage APIs, real-time data streams, quantitative signals, automated risk controls, position sizing, and performance monitoring.

EDUCATION

---

- **Dalhousie University** Halifax, Canada  
*Master of Applied Computer Science, GPA: 3.9/4.0* Jan 2021 – Aug 2022
- **University of Toronto** Toronto, Canada  
*Enterprise Architecture Certification (3-semester program)* Completed 2024
- **Gujarat Technological University** Ahmedabad, India  
*B.E. Computer Engineering, GPA: 3.92/4.0* Aug 2016 – Aug 2020